

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular and Supplementary Winter Examination – 2024

Course: B. Tech.

Semester : V

Branch : Electronics and Telecommunication Engineering/Electronics Engineering

Subject Code & Name: Digital Signal Processing BTETC502_BTEXC502

Max Marks: 60

Date: 08/02/2025

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Objective type questions. (Compulsory Question)

12

1 Why is quantization necessary in DSP?

CO1

1

- a) To improve resolution
- b) To convert to binary
- c) To reduce noise
- d) To amplify signals

2 Which of the following is NOT a requirement for DSP?

CO1

1

- a) Data Sampling
- b) Quantization
- c) Signal Compression
- d) Modulation

The DTFT of $\delta[n]$ (unit impulse) is:

CO2

1

- a) 0
- b) 1
- c) $e^{j\omega}$
- d) 2π

4	Parseval's theorem in DTFT relates:	CO2	1
	a) Energy in time and frequency b) Power in time and frequency c) Periodicity of signals d) Linearity of signals		
5	Scaling in the time domain corresponds to _____ in the frequency domain?	CO2	1
	a) Convolution b) Compression c) Expansion d) Time Scaling		
6	Which method can be used to compute linear convolution using circular convolution?	CO2	1
	a) Zero-padding b) Truncation c) Windowing d) Scaling		
7	The main advantage of FFT over DFT is:	CO4	1
	a) Higher accuracy b) Reduced computational complexity c) Reduced memory requirements d) Simplicity		
8	The Z-transform is particularly useful for analyzing:	CO3	1
	a) Continuous systems b) Discrete systems c) Analog systems d) Random systems		

9	For Z transform, the ROC must:	CO3	1
	a) Contain infinity b) Contain the unit circle c) Be bounded d) Exclude all poles		
10	Bilinear transformation introduces:	CO5	1
	a) Aliasing b) Frequency warping c) Phase distortion d) Oscillations		
11	The inverse Z-transform is used to:	CO3	1
	a) Find the Z-domain representation b) Convert a signal back to the time domain c) Compute the Fourier transform d) Find the region of convergence		
12	In multirate signal processing, an upsampler is used to:	CO5	1
	a) Increase the sampling rate b) Decrease the sampling rate c) Increase the frequency of the signal d) Decrease the frequency of the signal		
Q. 2 Solve the following.			
	A) Write the applications of Digital Signal Processing.	CO1	6
	B) Explain the term aliasing with neat diagrams. Explain various cases: under-sampling, over-sampling and normal sampling. What can be done to combat the effect of aliasing?	CO1	6
Q. 3 Solve the following.			
	A) State and prove any four properties of DTFT.	CO2	6
	B) Determine the DTFT of the signals:	CO2	6
	i) $x(n) = \sin(w_0 n)u(n)$ ii) $x(n) = \cos(w_0 n)u(n)$		

Q. 4 Solve Any Two of the following. 12

A) i) Compute the 4-point DFT of $x(n) = \{1, -1, 1, -1\}$ for $0 \leq n \leq 3$ CO2 6

ii) Compute the 4-point IDFT of $X(k) = \{0, -2j, 0, 2j\}$ for $0 \leq k \leq 3$

B) Use overlap –save method to convolve the following sequences: CO3 6

$x(n) = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ for $0 \leq n \leq 9$

and $h(n) = \{1, 0, -1\}$ for $0 \leq n \leq 2$
 Draw the sample ROC corresponding to the following system characteristics: CO3 6

- i) Causal and stable
- ii) Causal and unstable
- iii) Anti-causal and stable
- iv) Anti-causal and unstable
- v) Non-causal and stable
- vi) Non-causal and unstable

Q.5 Solve Any Two of the following. 12

A) Determine the Z- Transform and ROC of the following signals: CO3 6

- i) $x(n) = \{1, 2, 3, 4\}$ for $0 \leq n \leq 3$
- ii) $x(n) = \{1, 2, 3, 4\}$ for $-1 \leq n \leq 2$
- iii) $x(n) = \{1, 2, 3, 4\}$ for $-3 \leq n \leq 0$
- iv) $x(n) = \delta(n)$
- v) $x(n) = u(n)$
- vi) $x(n) = a^n u(n)$

B) Derive expression of $y(n)$ and draw the direct form- I and direct form- II realizations of the transfer function given by CO4 6

$$H(z) = \frac{1 + 2z^{-1} - z^{-2}}{1 + z^{-1} - z^{-2}}$$

C) Explain the concept of an ideal filter. What are the key requirements for a filter to be considered ideal? CO4 6

Q.6 Solve Any Two of the following. 12

A) If $H(z)$ was designed using bilinear transformation, determine its parent causal analog transfer function CO5 6

$$H(z) = \frac{4(z^2 + 3z + 4)}{10z^2 + 4z + 6}$$

B) Derive the expression to obtain the order of a Butterworth filter. CO5 6

C) Explain the applications of multirate digital signal processing. CO5 6

*** End ***